Name:_____ ID:_____

Q1	Q2	Q3	Q4	Q5	Q6	Q7	SUM
/15	/15	/14	/20	/15	/15	/16	/100

CSE 2046 Analysis of Algorithms

Spring 2020

Online Makeup Homework

Solution and Submission Notes

- **Textbooks, slides, and notes are open.**
- **Internet usage is not allowed; however, computers and smartphones may be used for viewing course materials provided for this course.**
- **Write your name, surname and student ID on the top of each exam solution page.**
- **Show all your work.**
- **Write the following sentence on the top of the first page with your handwriting and sign it: "On my honor, I have neither given nor received any unauthorized assistance on this examination."**
- **Either print this PDF file (6 pages) or use blank A4 pages for your solutions. Solve all the questions with your handwriting.**
- **Scan all the solution pages to a single PDF file named "firstname_lastname.pdf". Do not include raw photos, use an Android/IOS scanner application. All the writings should be very readable. Page layouts should be portrait, not landscape.**
- **Upload the PDF file to https://ues.marmara.edu.tr system before the deadline. Give at least 15-20 minutes for scanning and upload process.**

Name:_____ ID:_____

Q-1. (15 pts) Given the following DNA string:

G C A T C G A G C A G A G C A G A G C A G A G T

And the problem is to find ALL OCCURRENCES of the following pattern in the above DNA string: (Do not stop after finding one. Continue to find the others if any)

G A G C A G A G

(a – 7 pts) Apply Horspool algorithm to solve this problem. Do not forget to show the shift table. How many character comparisons will be performed?

(b – 8 pts) Repeat (a) for Boyer-Moore algorithm. (Show bad-symbol and good-suffix tables).

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Q-2. (15 pts) Remember the optimal binary search tree (BST) problem: Given n keys $a_1 < \ldots < a_n$ and probabilities $p_1 \leq \ldots \leq p_n$ searching for them, find a BST with a minimum average number of comparisons in successful search.

Apply dynamic programming algorithm for the Optimal BST problem for keys A, B, C and D with search probabilities 0.25, 0.4, 0.15, and 0.2, respectively.

Name:_____ ID:_____

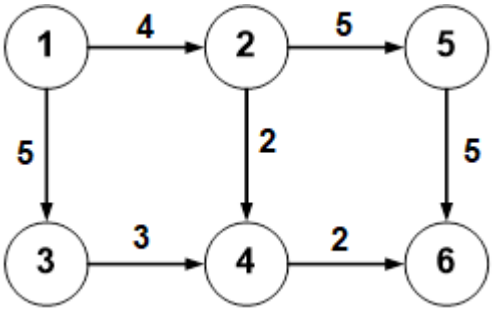
Q-3. (14 pts) Suppose we have a file that contains 11 *a*, 16 *e*, 12 *i*, 3 *s*, 4 *t*, 13 blanks, one 0 and one *newline*.
(a – 11 pts) Using the Huffman algorithm compute the optimal coding for each character.

(b – 3 pts) How much would be the size reduction of the file compared to an ASCII coded version?

Name:_____ ID:_____

Q-4. (20 pts) (a – 12 pts) Apply the <u>shortest augmenting path algorithm</u> to find a <u>maximum flow</u> and a <u>minimum cut</u> in the following network.

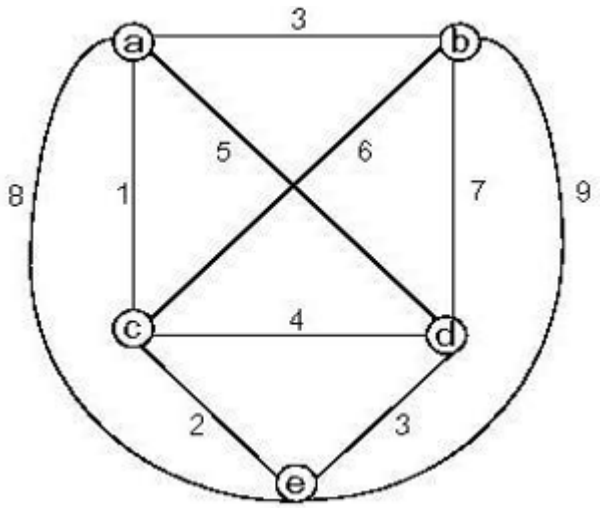
While applying BFS, visit the neighbor with smaller ID first. Show the queue and the labels clearly.



(b – 8 pts) After finding a maximum flow f^* in a graph, capacity of one of the edges (let's say edge (u,v)) is reduced by one. You do not want to calculate the max flow from the beginning. Describe a fast algorithm to find the new maximum flow.
Note: This part is not related to the above graph. Question is general.

Name:_____ **ID:**_____

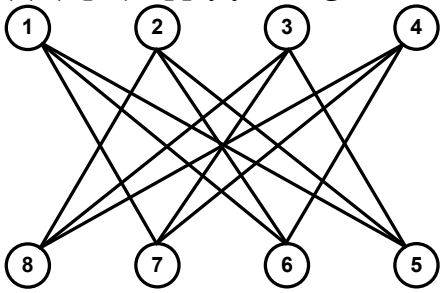
Q-5. (15 pts) Consider a **modified version of Traveling Salesman problem**. Again there are *n* cities and cost values between cities. The traveling salesman should start from a specified city (say city *a*) and he needs to visit all cities EXCEPT ONE, and go back to city *a*. The aim is to find the tour (of length *n*-1) with minimum total cost.
Define a bounding function and apply branch and bound algorithm for this modified version of TSP problem on the following graph that we have seen in the class. (Note: You will get a zero grade if you solve ordinary TSP problem.)



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Q-6. (15 pts) Consider the Graph Coloring Problem which seeks to assign the smallest number of colors to the vertices of a graph so that no two adjacent vertices are the same color.
(a) (8 pts) Describe an approximation algorithm for Graph Coloring Problem. (Do not seek from the Internet. Write your own algorithm.)

(b) (7 pts) Apply your algorithm in the following graph. What is the accuracy ratio?



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Q-7. (16+10 pts)
(a – 7 pts) Using decision-tree approach, find a lower bound for finding the maximum of *n* numbers by comparison-based algorithms? Is this bound tight? Why?

(You do not have to draw the decision tree. You can just describe how you find the lower bound.)

(b – 9 pts) Suppose that Problem X can be reduced to Minimum Spanning Tree problem and Problem Y can be reduced to Knapsack problem. All reductions can be done in linear (O(n)) time. For each of the following, indicate whether it is **True, False, or Unknown. Give your reasoning, otherwise your answer will not be graded.**

(i) Problem Y is more difficult than Problem X.

(ii) Knapsack problem can be reduced to Problem X in linear time.

(iii) Both problem X and Problem Y can be tractable problems.

(Bonus question – 10 pts) Prove that decision version of the problem given in Question 5 (modified TSP) is also NP-Complete.